

Lecture/Week 14Numerical Solutions of ODEsNumerical Methods for Solution of First Order ODEs

- a) Euler's Method b) Improved Euler's Method
 c) 2nd Order RK Method d) 4th Order RK Method.

Given i) First Order ODE. $\frac{dy}{dx} = f(x, y)$

ii) Initial condition $y(x_0) = y_0$

iii) Domain

$$x_0 \leq x \leq x_n$$

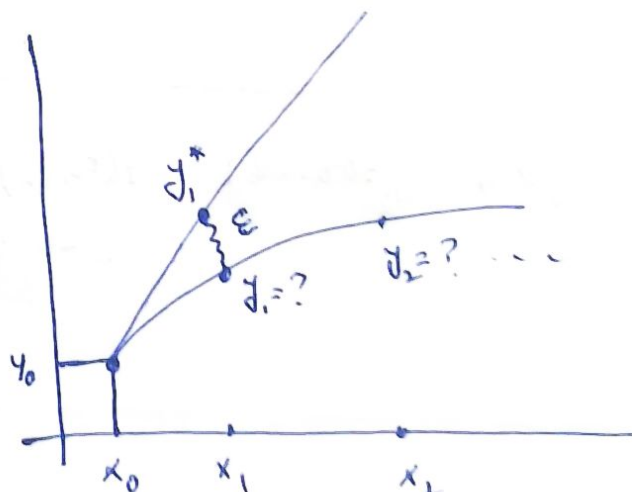
iv) step size

$$h = \frac{b-a}{n} = \frac{x_n - x_0}{n}$$

a) Euler's Method

$$y_{n+1} = y_n + h f(x_n, y_n)$$

Formula ↗



Example $\frac{dy}{dt} + 2y = 2 - e^{-4t}$; $y(0) = 1$ (2)

Use Euler's method with $h=0.1$ and compute the solution of ODE at $t = 0.1, 0.2, 0.3, 0.4, 0.5$

Solution $\frac{dy}{dt} = 2 - 2y - e^{-4t}$

$f(t, y) = 2 - 2y - e^{-4t}$; $t_0 = 0$; $y_0 = 1$; $h = 0.1$

Domain $0 \leq t \leq 0.5$

Formula :- $y_{n+1} = y_n + h f(t_n, y_n)$

n	t_n	y_n	$f(t_n, y_n)$	$y_{n+1} = y_n + h f(t_n, y_n)$
0	$t_0 = 0$	$y_0 = 1$	$f(t_0, y_0) = f(0, 1)$ $= -1$	$y_1 = y(0.1) = y_0 + h f(t_0, y_0)$ $= 1 + 0.1(-1) = 0.9$
1	$t_1 = 0.1$	$y_1 = 0.9$	$f(t_1, y_1) = f(0.1, 0.9)$ $= -0.470320$	$y_2 = y(0.2) = y_1 + h f(t_1, y_1)$ $= 0.9 + (0.1)(-0.470320)$ $= 0.852967$
2	$t_2 = 0.2$	$y_2 = 0.852967$	$f(t_2, y_2) = f(0.2, 0.852967)$ $= -0.1552649$	$y_3 = y_2 + h f(t_2, y_2)$ $= 0.852967 + (0.1)(-0.1552649)$ $= 0.8374415$
3	$t_3 = 0.3$			

Improved Euler's Method

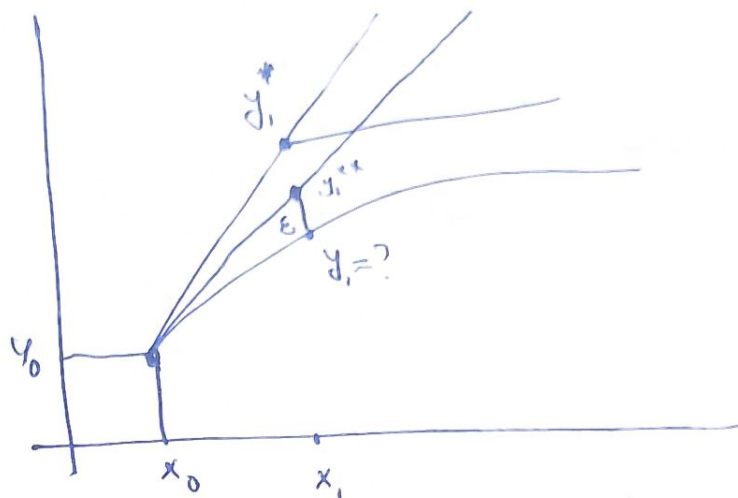
Modified Euler's Method (Heun's Method)

$$\frac{dy}{dx} = f(x, y) \rightarrow (1)$$

$$y(x_0) = y_0 = (x_0, y_0)$$

$$m_1 = \left. \frac{dy}{dx} \right|_{(x_0, y_0)} = f(x_0, y_0)$$

$$\boxed{m_1 = f(x_0, y_0)}$$



Equation of straight line through (x_0, y_0) with slope m_1
 $y - y_0 = m_1(x - x_0)$

$$\boxed{y - y_0 = f(x_0, y_0)(x - x_0)}$$

When $x = x_1$, $y = y_1^*$

$$\boxed{y_1^* = y_0 + h f(x_0, y_0)}$$

Now we have point (x_1, y_1^*)

$$m_2 = \left. \frac{dy}{dx} \right|_{(x_1, y_1^*)} = f(x_1, y_1^*)$$

Average slope $m = \frac{m_1 + m_2}{2} = \frac{1}{2} [f(x_0, y_0) + f(x_1, y_1^*)]$

Eg. of line through (x_0, y_0) with slope m .

$$\boxed{y - y_0 = \frac{1}{2} [f(x_0, y_0) + f(x_1, y_1^*)] (x - x_0)}$$

When $x = x_1$ and $y = y_1^{**}$

$$y_1^{**} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^*)]$$

Now Improved Euler's Method

$$y_1 \approx y_1^{**}$$

Therefore

$$y_1 = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^*)]$$

$$\text{where } y_1^* = y_0 + h f(x_0, y_0)$$

Now using Induction

$$y_2 = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^*)]$$

$$y_2^* = y_1 + h f(x_1, y_1)$$

General formula

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^*)]$$

$$y_{n+1}^* = y_n + h f(x_n, y_n)$$

(5)

Example

$$y' = y - x$$

$$y(0) = \frac{1}{2}$$

Use modified Euler method with $h=0.1$ to approximate the solution at $x=0.5$

Solution

$$f(x, y) = y - x ; x_0 = 0 ; y_0 = \frac{1}{2} ; h = 0.1$$

$$0 \leq x \leq 0.5$$

Formula :-
$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^*)]$$

$$y_{n+1}^* = y_n + h f(x_n, y_n)$$

n	x_n	y_n	$f(x_n, y_n)$	x_{n+1}	y_{n+1}^*	$f(x_{n+1}, y_{n+1}^*)$
0	$x_0 = 0$	$y_0 = \frac{1}{2}$	$f(x_0, y_0) = f(0, \frac{1}{2}) = \frac{1}{2}$	$x_1 = 0.1$	$y_1^* = y_0 + h f(x_0, y_0) = \frac{1}{2} + (0.1)(\frac{1}{2}) = 0.75$	$f(x_1, y_1^*) = f(0.1, 0.75) = 0.65$

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^*)]$$

$$y_1 = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^*)]$$

$$= 0.5 + \frac{0.5}{2} [0.75 + 0.65] = 0.85$$

1. $x_1 = 0.1$ ~~$y_1 = 0.85$~~ $y_1 = 0.85$ $f(x_1, y_1) = f(0.1, 0.85) = 0.75$ $x_2 = 0.2$ $y_2^* = y_1 + h f(x_1, y_1) =$

Example

$$y' = t + \sqrt{y} \quad y(0) = 1$$

$$h = 0.2 \quad 0 \leq t \leq 0.6$$

Solution

$$f(t, y) = t + \sqrt{y}; \quad t_0 = 0; \quad y_0 = 1; \quad h = 0.2$$

$$0 \leq t \leq 0.6$$

n	t_n	y_n	$f(t_n, y_n)$	x_{n+1}	y_{n+1}^*	$f(x_{n+1}, y_{n+1}^*)$
0	$t_0 = 0$	$y_0 = 1$	$f(t_0, y_0) = f(0, 1) = 1$	$x_1 = 0.2$	$y_1^* = y_0 + h f(x_0, y_0)$ $= 1 + 0.2(1)$ $= 1.2$	$f(x_1, y_1^*) = f(0.2, 1.2)$ $= 1.295$

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^*)]$$

$$\begin{aligned} y_1 = y(0.2) &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^*)] \\ &= 1 + \frac{0.2}{2} [1 + 1.295] = \underline{\underline{1.2295}} \end{aligned}$$

1	$t_1 = 0.2$	$y_1 = 1.2295$	$f(t_1, y_1) = 1.30885$	$x_2 = 0.4$	$y_2^* = y_1 + h f(t_1, y_1)$ $= 1.2295 + 0.2(1.30885)$ $= 1.4913$	$f(x_2, y_2^*) = f(0.4, 1.4913)$ $= 1.62119$
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$$\begin{aligned} y_2 = y(0.4) &= y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^*)] \\ &= 1.2295 + \frac{0.2}{2} [1.30885 + 1.62119] \\ &= \underline{\underline{1.5225}} \end{aligned}$$

Second Order RK-Method

(7)

Given $\frac{dy}{dx} = f(x, y)$ $y(x_0) = y_0$

Taylor Expansion

$$y(x+h) = y(x) + h y'(x) + \frac{h^2}{2!} y''(x) + \dots \rightarrow (1)$$

$$\text{Now } y''(x) = \frac{d}{dx} (y'(x)) = \frac{d}{dx} (f(x, y))$$

$$= f_x(x, y) + f_y(x, y) \frac{dy}{dx}$$

$$y''(x) = f_x(x, y) + f_y(x, y) f(x, y) \rightarrow (2)$$

use (2) in (1)

$$y(x+h) = y(x) + h f(x, y) + \frac{h^2}{2} \left[f_x(x, y) + f_y(x, y) f(x, y) \right]$$

$$= y(x) + \frac{h}{2} f(x, y) + \frac{h}{2} f(x, y) + \frac{h^2}{2} \left[f_x(x, y) + f_y(x, y) f(x, y) \right]$$

$$= y(x) + \frac{h}{2} f(x, y) + \frac{h}{2} \left[f(x, y) + h f_x(x, y) + h f_y(x, y) f(x, y) \right] \rightarrow (3)$$

We know

$$f(x+h, y+h) = f(x, y) + h \left[f_x(x, y) \right] + k \left[f_y(x, y) \right] \rightarrow (4)$$

Comparing (3) and (4)

$$y(x+h) = y(x) + \frac{h}{2} f(x, y) + \frac{h}{2} f(x+h, y+h f(x, y))$$

$$y_{n+1} = y_n + h \left[\frac{k_1}{2} + \frac{k_2}{2} \right]$$

$$k_1 = f(x_n, y_n) \quad \text{and} \quad k_2 = f(x_{n+1}, y_n + h f(x_n, y_n))$$

Fourth Order R-K Method

(8)

$$y' = f(x, y) \quad y(x_0) = y_0$$

$$y_{n+1} = y_n + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

where

$$k_1 = h f(x_n, y_n)$$

$$k_2 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_{n+1}, y_n + k_3) \text{ or } h f(x_{n+1}, y_n + k_3)$$

(9)

Example

$$y' = -2x - y \quad y(0) = -1$$

Use 2nd Order RK-method to find $y(0.2)$ with $h = 0.1$

Solution $f(x, y) = -2x - y$; $x_0 = 0$; $y_0 = -1$; $h = 0.1$

$$0 \leq x \leq 0.2$$

Formula $y_{n+1} = y_n + h \left[\frac{k_1}{2} + \frac{k_2}{2} \right]$

$$k_1 = f(x_n, y_n)$$

$$k_2 = f(x_{n+1}, y_n + hk_1) \\ = f(x_{n+1}, y_n + hf(x_n, y_n))$$

<u>n</u>	<u>x_n</u>	<u>y_n</u>	<u>$k_1 = f(x_n, y_n)$</u>	<u>$y_n + hk_1$</u>	<u>$k_2 = f(x_{n+1}, y_n + hk_1)$</u>
0	$x_0 = 0$	$y_0 = -1$	$k_1 = f(x_0, y_0)$ $= f(0, -1)$ $= 0.1$	$y_0 + hk_1$ $= -1 + (0.1)(0.1)$ $= -0.99$	$k_2 = f(x_1, y_0 + hk_1)$ $= f(0.1, -0.99)$ $= 0.79$

$$y_{n+1} = y_n + h \left[\frac{k_1}{2} + \frac{k_2}{2} \right]$$

$$y_1 = y(0.1) = y_0 + 0.1 \left[\frac{0.1}{2} + \frac{0.79}{2} \right] = -1 + (0.1)(0.895) \\ = -0.9105$$

1	$x_1 = 0.1$	$y_1 = -0.9105$	$k_1 = f(x_1, y_1)$ $= f(0.1, -0.9105)$ $= 0.7105$	$y_1 + hk_1$ $= -0.9105 + (0.1)(0.7105)$ $= -0.83945$	$k_2 = f(x_2, y_1 + hk_1)$ $= f(0.2, -0.83945)$ $= 0.43945$
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$$y_2 = y(0.2) = y_1 + h \left[\frac{k_1}{2} + \frac{k_2}{2} \right] \\ = -0.9105 + 0.1 \left[\frac{0.7105}{2} + \frac{0.43945}{2} \right]$$

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